A Static Voltage Stability Enhancement Method for Power Systems With a High Proportion of Voltage Source Converters

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***Abstract*—One of the main characteristics of power systems with a high proportion of renewable energy is that the power system is becoming a power system with a high proportion of voltage source converters (VSCs). Owing to the current limitation of VSCs, the reactive power regulation range is limited, thus greatly weakening the effective reactive power reserve and reducing the static voltage stability. In this paper, the PQ regulation characteristics of the VSC are first analyzed considering the current limitation. By establishing the PQ characteristic analytical model of a single grid-connected VSC system, the changes in the system reactive power reserve and system static voltage stability evolution law under different penetration ratios of VSCs are analyzed. On this basis, according to the proposed reactive power support sensitivity of each source bus, it is proposed to optimally determine the voltage and effective reactive power reserve of each source bus to realize effective enhancement of the static voltage stability margin of the power system with a high proportion of VSCs. Simulation analyses are carried out in a system with multiple grid-connected VSCs and an extended IEEE-39 bus system, which validate the effectiveness and accuracy of the proposed method.**

***Index Terms*—Current limitation, effective reactive power reserve, high proportion of voltage source converters, reactive power support sensitivity, static voltage stability.**

I. Introduction

THE deployment of renewable energy such as wind power and photovoltaics, which are connected to the grid via inverters, is increasing worldwide to meet decarbonization targets. Renewable energy is the backbone of the shift from fossil fuels to clean and net zero emission systems [1], [2]. In this context, many synchronous generators are replaced by renewable energy, and direct-driven wind and photovoltaic power sources with voltage source converters (VSCs) as grid-connected interfaces become the main power sources. However, the current limitation of VSCs results in a smaller reactive power regulation range than that of a synchronous generator of the same capacity. This will cause many new problems in the safe and stable operation of the system, one of which is system static voltage stability [3], [4], [5].

The connection of VSCs can significantly impact the voltage conditions of consumers and utility equipment, especially under high penetration levels. These impacts can be positive or negative depending on the system operating characteristics and the VSC itself [6], [7]. The impact of VSCs on the voltage stability phenomenon is an important issue that needs investigation.

Voltage instability is characterized by a progressive de-crease in voltage magnitude and occurs essentially because of the system inability to meet a growing demand for reactive power at certain buses in stressed situations [8]. Voltage instability is usually of slow nature (minutes or hours), and is driven by the action of discrete type devices and load variations [9]. The static voltage instability phenomenon is not new.

However, most studies address the impact of VSCs on short-term voltage stability and fault-ride-through requirements [10], [11], [12]. In [13], their study focused on current limitation for voltage-controlled inverters during overloads caused by poor transient load sharing between inverters and synchronous generators in islanded microgrids. In [14], a two-stage simultaneous control scheme was proposed for improving the transient stability of a VSC while considering the current limitation during the fault state and voltage support after fault clearance. To the best of the authors' knowledge, the current literature still lacks a wider and detailed analysis considering the current limitation impacts of VSC on static voltage stability with a detailed representation of system dynamic. Furthermore, this paper focuses only on static voltage stability because of the high proportion of VSCs, and transient voltage problems during fault conditions are beyond the scope of the proposed model.

Many studies have been carried out on the problem of static voltage stability with a high proportion of VSCs [15], [16], [17]. However, most studies have focused on constructing stability indicators and analyzing the change in static voltage stability with increasing penetration ratio. In [18], on the basis of Norton

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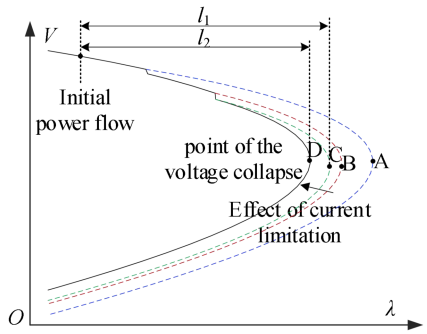


Fig. 1. P-V curve considering the current limitation of VSC.

current distribution characteristics, a node static voltage stability index named the H index was constructed, which was adapted for wind farms. In [19], the load conductivity of the distribution network was constructed to study the static voltage stability change under different penetration ratios of distributed photovoltaics. For the impact of different PCCs of VSCs on static voltage stability, in [20], the influence of the penetration of wind power, different points of common coupling of wind farms, load growth patterns and other factors on static voltage stability was analyzed. Overall, these results are based on the current operating status. However, the impact of the current limitation on the static voltage stability problem has received less attention.

When the voltage of the points of common coupling of the VSC is continuously low, the current dispatched by the VSC will reach the limit value. Moreover, the reactive power dispatched by the VSC decreases with decreasing reduction. When the system approaches the point of the voltage collapse, the limitation characteristic of the VSC will form a positive feedback loop of the voltage drop-reactive power drop, further deteriorating the system static voltage stability. The P-V curve reflecting the impacts of the current limitation is shown in the Fig. 1.

Because of the current limitation, the point of voltage collapse changes from point D to point C, and the distance from the initial power flow to the point of voltage collapse is shortened from l_1 to l_2. However, this impact has not yet been quantified.

It is well known that effective reactive power reserve optimization can improve system stability without adding any other reactive power resources, since reactive power support is strongly related to voltage instability problems. In terms of reactive power limit of VSC, in [21], [22], current limits are used to define the active and reactive power limits of VSC. However, no basis is given for the setting of active and reactive current limits. Recently, the interconnection and interoperability of VSC standards such as IEEE Std 1547-2018 [23], IEEE Std 2800-2022 [24], GB/T 19964-2024 (China) [25], and GB/T 36547-2024 (China) [26] have been drafted, in [23], [24], [25], [26], the minimum range for the reactive power capability of inverter-based resources is required. Based on the standards, we can get the reactive power capability shall be met for all active power output levels (including at zero) of the VSC.

The method keeps more reactive power reserves [27], [28], [29] of the generator, the reactive power weight of which is larger. One previous study [30] relied on the concept of effective reactive power reserves and evaluates the impact of generators on the maximum permissible loading power. Another study [31] explored the underaddressed topic of reactive power reserve during the energy transition. In [32], [33], the sensitivity of the reactive power dispatched by the synchronous generator bus to the reactive power growth of the load bus was measured.

The above study investigated the value of the reactive power reserve for a synchronous generator in detail. However, owing to the current limitation of VSCs, the traditional reactive power weight is not suitable for systems with a high proportion of voltage source converters. The influence on the system reactive power reserve of grid-connected VSCs will need to consider the bus voltage immunity.

The contributions of this paper are summarized as follows:

1) To the best of the authors' knowledge, the impact of VSC current limitation on static voltage stability has not been well studied. Many of the new components based on power electronics will require a dynamic representation to accurately represent limits. However, proper modeling of traditional steady-state conditions is also important. The derivations of the VSC limitations are therefore useful. Accordingly, the authors attempt to analyze the effect of VSC current limitation on the static voltage stability.
2) In this paper, a PQ characteristic analytical model of a single grid-connected VSC system considering the current limitation is first established, and the influencing mechanism of the current limitation on the static voltage stability domain of the system is analyzed.
3) By analyzing the PQ regulation characteristics of the VSC and considering the ability to maintain the static voltage stability and voltage immunity of generator bus, the reactive power support sensitivity is proposed.
4) It is proposed to optimally determine the voltage and reactive power of generator bus in base case. The generator bus with a large reactive power support sensitivity will retain a more effective reactive power reserve. So, the static voltage stability margin of the system with a high proportion of VSCs can be improved more effectively.

The remainder of the paper is organized as follows. Section II introduces the reactive power regulation range and current limitation of VSCs. Section III presents the static voltage stability analysis of a single grid-connected VSC system. Section IV introduces the static voltage stability enhancement method for a power system with a high proportion of VSCs. Simulations are provided in Sections V, and VI concludes the paper.

II. Analysis of the PQ Characteristics of VSCs

A. Model of the PQ Characteristics of the VSC

To facilitate the unification of the research objects, this paper restricts the discussion only to full-scale converters. If the current is too high, damage may occur to the semiconductors in converts. Any control designed to operate under abnormal grid conditions must adhere to the physical current limitation of the semiconductors, and a limit is set for the converter current. Therefore, a VSC is usually connected to a grid via the control strategy shown in Fig. 2 [34].

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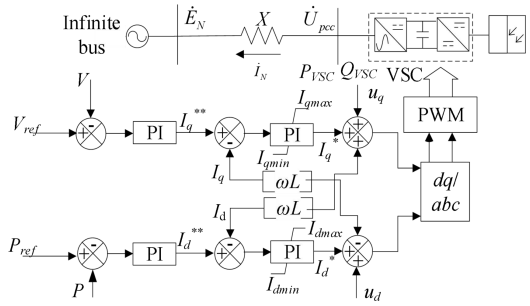


Fig. 2. The constant V control strategy of the VSC.

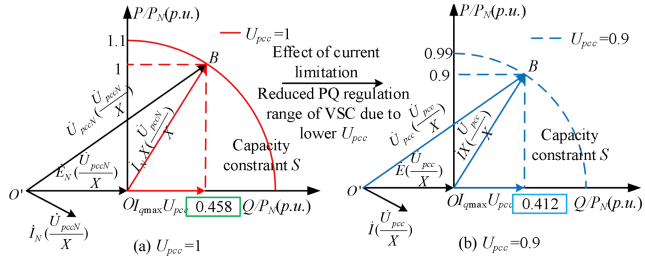


Fig. 3. P-Q curve of the VSC.

The maximum current limitation is usually based on the d-axis current priority limitation or the dq-axis current proportional limitation. To provide both active power and reactive power support, this paper uses the dq-axis axis current proportional limit. The dq-axis output current is as follows:

$$I_d^* = \begin{cases} I_d^{**} & I_d \leqslant I_{d\max} = k_{dq} I_{\max} \\ I_{d\max} & I_d > I_{d\max} = k_{dq} I_{\max} \end{cases} \tag{1}$$

$$I_q^* = \begin{cases} I_q^{**} & I_q \leqslant I_{q\max} = \sqrt{1-k_{dq}^2} I_{\max} \\ I_{q\max} & I_q > I_{q\max} = \sqrt{1-k_{dq}^2} I_{\max} \end{cases} \tag{2}$$

where $I_{\max}$ is the inverter current limit value, $I_d{}^{**}$ is the d-axis current command, $I_q{}^{**}$ is the q-axis current command and k_{dq} is the proportional value.

$I_{d\max}$ is the rated d-axis current I_{dN}, which is the d-axis current at the rated active power, rated voltage and rated power factor. The long-term steady state current $I_{\max}$ is determined by capacity. $I^2{}_{\max} = I^2{}_{d\max} + I^2{}_{q\max}$, so we can obtain the value of $I_{q\max}$. The VSC can operate in- four quadrants, $I_{d\min} = -I_{d\max}$, and $I_{q\min} = -I_{q\max}$. While Photovoltaic and direct-drive wind turbines are not considered to absorb active power, $I_{d\min}$ is set as $I_{d\min} = 0$.

In Fig. 2, on the basis of the grid voltage vector $\dot{U}_{pcc}$ oriented, where $u_q = 0$ and $u_d = U_{pcc}$, $P_{\rm VSC}$, and $Q_{\rm VSC}$ are numerically shown in (3) and (4) [35]:

$$P_{\rm VSC} = u_d I_d^* \tag{3}$$

$$Q_{\rm VSC} = u_d I_q^* \tag{4}$$

According to (3) and (4), when the dq-axis current command values $I^*{}_q$ and $I^*{}_d$ are constant, the active and reactive power dispatched by the VSC is proportional to the voltage of the PCC, which is highly susceptible to voltage perturbations. Notably, regardless of whether the VSC adopts the constant V grid-connected control strategy or other grid-connected control strategies such as constant Q, constant I or constant power factor and regardless of whether the VSC adopts a grid-forming or grid-following control strategy, when the current reaches the maximum limitation value, the VSC presents the current source characteristic.

In [21], [22], the power limit is set as follows:

$$P_{\rm VSC\,max} = u_{dN} I_{d\,\max}^* = k_{dq} S_{\rm VSC} \tag{5}$$

$$Q_{\rm VSC\,max} = u_{dN} I_{q\,\max}^* = \sqrt{1-k_{dq}^2} S_{\rm VSC} \tag{6}$$

where $P_{\rm VSCmax}$ is the maximum active power dispatched by the VSC, $Q_{\rm VSCmax}$ is the maximum reactive power dispatched by the VSC, and $S_{\rm VSC}$ is the capacity of the VSC. Since IEEE Std 1547-2018 is used for distributed energy resources, it is not suitable for the transmission electric power systems. In IEEE Std 2800-2022, minimum reactive power injection and absorption capability is $0.3287{*}P_{\rm VSCN}$, which equals to $0.313{*}S_{\rm VSC}$. In GB/T 19964-2024 and GB/T 36547-2024, the minimum reactive power injection and absorption capability is $0.436{*}S_{\rm VSC}$. To fully utilize the converter capacity, the rated capacity of VSCs, such as photovoltaic and direct-drive wind turbines, which are connected to the grid with full-scale converters, is usually set to 1.1 times the rated power [36]. So, $I_{\max} = 1.1{*}I_{d\max}$. Therefore, a typical value of k_{dq} can be taken as 0.909. which also met the requirement of minimum range for the reactive power capability of VSC in above standards.

B. Comparison of the PQ Output Characteristics and Regulating Capabilities of the VSC and Synchronous Generator

On the basis of the grid-connected system shown in Fig. 2, the relationship between the voltage of the PCC and the grid voltage is as follows:

$$\dot{U}_{pccN} = \dot{E}_N + j\dot{I}_N X \tag{7}$$

According to (1)–(7), the phasor diagram and PQ curves of the VSC are shown in Fig. 3, assuming that all phases in Fig. 3 are multiplied by U_{pccN}/X, where the length of the OB in Fig. 3 represents the rated apparent power $S_{\rm VSC}$ of the VSC.

Fig. 3 shows that, due to the existence of the current limitation, when U_{pcc} is continuously low, the current dispatched by the VSC will reach the limit value. Moreover, the power dispatched by the VSC decreases with the reduction in U_{pcc}.

On the basis of the operating constraints of the synchronous generator described in [37], the PQ curve of the synchronous generator can be obtained as shown in Fig. 4 by considering other constraints, such as the maximum stator current and maximum rotor current limits, prime machine power constraints and stator end temperature rise constraints.

All phasors in Fig. 4 are assumed to be multiplied by $U_{\rm GN}/x_d$, where x_d is the generator d-axis reactance. As we can see in the Fig. 4, a synchronous generator can only reach its rated value of apparent power and its capacity can be most fully utilized when it is operated at rated voltage, current and power factor. Therefore, the rated reactive power is usually taken as the maximum reactive power for synchronous generators. The power factor $\cos\varphi_N$ of

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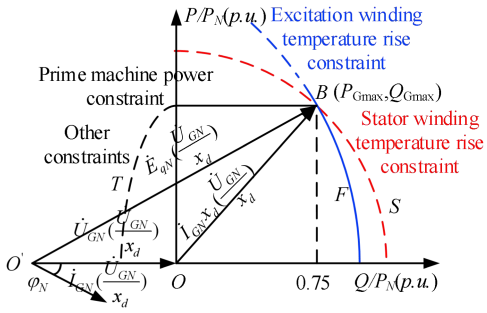


Fig. 4. P-Q curve of the synchronous generator.

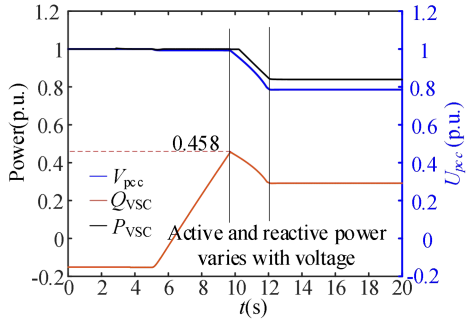


Fig. 5. The active and reactive power characteristic curves of the VSC.

most synchronous generators is set to K_1. We can obtain:

$$Q_{\mathrm{VSCmax}} = \sqrt{1-k_{dq}^2}S_{\mathrm{VSC}} \tag{8}$$

$$Q_{\mathrm{G\,max}} = \sqrt{1-K_1^2}S_{\mathrm{G}} \tag{9}$$

Since the penetration ratio of the VSC is defined as the active power dispatched by the VSC/the total active power outputs, When $P_{\mathrm{VSCmax}} = P_{\mathrm{Gmax}}$, the maximum reactive power of the VSC with the maximum reactive power of the synchronous unit, the following equation holds:

$$Q_{\mathrm{VSCmax}} = \frac{\sqrt{1-k_{dq}^2}K_1}{\sqrt{1-K_1^2k_{dq}}}Q_{\mathrm{G\,max}} \tag{10}$$

The generator excitation winding consists of mechanical equipment such as copper wire, which can withstand a long period of stable operation of the excitation current and the value of K_1 can be selected between 0.8-0.85.

C. Analysis of the PQ Regulation Capability of the VSC

The 200 MVA grid-connected direct-drive wind farm using the constant V control strategy shown in Fig. 2 is taken as an example. In the power flow equation, the bus which controls the terminal voltage instead of regulating the reactive power is regard as a PV bus, the reactive power of which can be calculated at the convergence of the power flow equation as shown in (11).

$$Q_i = U_i\sum_{j=1}^{j=n}U_j(G_{ij}\sin\delta_{ij} - B_{ij}\cos\delta_{ij}) \tag{11}$$

Where, U_i and U_j are the voltage amplitude of bus i and bus j; G_{ij} and B_{ij} are the real and imaginary parts of the admittance matrix element Y_{ij}, respectively; $\delta_{ij} = \delta_i$-δ_j, δ_i is the phase angle of bus voltage U_i, δ_j is the phase angle of bus voltage U_j.

The reactive power of VSC is not infinite, when the load is increasing continuously, the reactive power of VSC will reach the Q_{VSCmax}. When the reactive power output of VSC is lower than its maximum reactive power, the var output of VSC can be calculated based on (11), when the var output of VSC is higher than its maximum reactive power, the var output of VSC can be calculated based on (4). So, the var output of VSC can be included in the static voltage stability model.

The PI control parameters in Fig. 2 are chosen by root locus method, which is a typical method [38]. This paper focuses on the long-term stable output power characteristics of the VSC, and the optimization of the control parameters is not the focus of this research. Thus, the optimization of the PI parameters of the control strategy is not considered in this paper. The stability and control accuracy of the VSC control strategy are of greater concern in this paper, and the response speed is not a priority. Fig. 5 shows the PQ regulation characteristic curve of the wind farm after the current reaches the limit value in the PSCAD.

As shown in Fig. 5, after a load increase results in a decrease in U_{pcc}, to maintain a constant U_{pcc}, the wind farm gradually increases the reactive power outputs, and U_{pcc} remains constant until I_q reaches the limit value. At 9.8 s, I_q reaches the limit value $I_{q\mathrm{max}}$, and the reactive power dispatched by the wind farm decreases as U_{pcc} decreases, whereas the active current I_d of the wind farm then reaches the limit value $I_{d\mathrm{max}}$ at 10.2 s. The active power of the wind farm also decreases as U_{pcc} decreases, and the wind farm exhibits current source characteristics. This property complicates the static voltage stability problem as the system approaches the point of voltage collapse.

III. Static Voltage Stability Analysis of a Single Grid-Connected VSC System

For the static voltage stability problem of a single grid-connected VSC system, related studies have been performed [39], [40], and the stability mechanism has been clarified. On the basis of previous studies, the paper focuses on the reactive power regulation range and the current limitation of VSCs. The effects of multiple grid-connected VSCs are further analyzed in subsequent sections.

Depending on the PCC, the topology of a single grid-connected VSC system can be classified into two types, as shown in Fig. 6. In Fig. 6, X_L and R_L are the equivalent reactance and resistance of the load, respectively. It is assumed that the phasor angle of bus voltage V is 0 and the phasor angle of bus voltage E is δ. Thus, the apparent power consumed by the load is:

$$\dot{I} = \frac{\dot{E}}{Z_L + Z_S} \tag{12}$$

$$|I|^2 = \frac{E^2}{(Z_L + Z_S)^2} \tag{13}$$

$$S = |I|^2|Z_L| = \frac{E^2}{\left(\sqrt{R_L^2 + X_L^2} + X_S\right)^2}\sqrt{R_L^2 + X_L^2} \tag{14}$$

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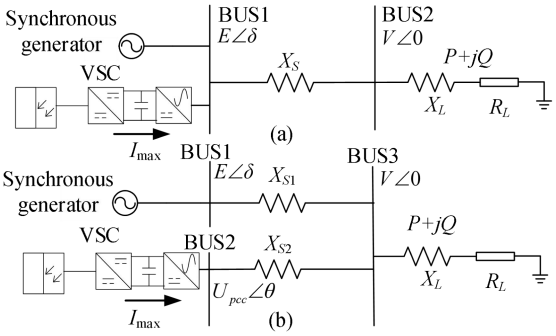


Fig. 6. Diagram of a single VSC grid-connected system. (a) The PCC of the VSC is the generator bus; (b) the PCC of the VSC is not the generator bus.

Where, Z_L is the load equivalent impedance; Z_S is the system equivalent impedance. Therefore, the maximum apparent power consumed by the load is obtained when the load equivalent impedance is equal to the system equivalent impedance:

$$\sqrt{R_L^2 + X_L^2} = X_S \tag{15}$$

For the system to be able to transmit the maximum apparent power, the system reactive power reserve Q_{res} must be reached:

$$Q_{\text{res}} = \frac{E^2}{4X_S^2}X_S + \frac{E^2}{4X_S^2}X_L = \frac{E^2}{4X_S} + \frac{E^2}{4X_S^2}X_L \tag{16}$$

When $X_L = X_S$, the maximum system reactive power reserve required is $E^2/2X_S$. To ensure that the system still transmits the maximum apparent power in either direction of load power growth, the effective system reactive power reserve is taken to be $E^2/2X_S$.

In Fig. 6(a), it is assumed that the reactive power reserve of the synchronous generator Q_{Gres} is $E^2/2X_S$ before the VSC replaces the synchronous generator. The voltage of BUS1 can be maintained constant until the reactive power reserve of BUS1 is consumed. The limit-induced bifurcation (LIB) occurs immediately after the reactive power reserve of BUS1 is exhausted, resulting in system voltage collapse. Hence, the voltage of BUS1 can remain constant throughout the load growth to the static voltage stability limit; therefore, the static voltage stabilization domain of the system shown in Fig. 6(a) is not affected by the current limitation.

There is a relationship between E and V:

$$\dot{E} = \dot{V} + \frac{Q}{V}X_S + j\frac{P}{V}X_S \tag{17}$$

Then, it can be obtained via (17):

$$E^2 = V^2 + 2QX_S + \frac{(P^2+Q^2)X_S^2}{V^2} \tag{18}$$

Therefore, by solving (18), we can obtain:

$$V^2 = \frac{(E^2 - 2QX_S)}{2} + \frac{1}{2}\sqrt{(E^2-2QX_S)^2 - 4(P^2+Q^2)X_S^2} \tag{19}$$

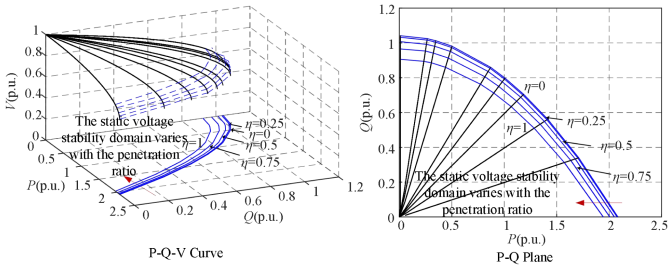


Fig. 7. The static voltage stability domain varies with the penetration ratio.

At the LIB, the line consumed reactive power and load consumed reactive power are equal to Q'_{G}:

$$\frac{P^2+Q^2}{V^2}X_s + Q = Q'_{\text{G}} \tag{20}$$

In (20), Q'_{G} is as follows:

$$Q'_{\text{G}} = Q_{\text{VSCres}} + Q_{\text{Gres}} = \frac{\eta\sqrt{1-k_{dq}^2}K_1}{\sqrt{1-K_1^2k_{dq}}}\frac{E^2}{2X_S} + (1-\eta)\frac{E^2}{2X_S} \tag{21}$$

where η is the penetration ratio of the VSC; Q_{VSCres} represents the reactive power reserves of VSC; Q_{Gres} represents the reactive power reserves of synchronous generator.

Assuming that $Q = kP$ between the load reactive power Q and active power P, k is the direction of load growth. Bringing (19) and (21) into (20):

$$\frac{(1+k^2)P^2}{\frac{E^2}{2X_S} - kP + \sqrt{\left(kP - \frac{E^2}{2X_S}\right)^2 - (1+k^2)P^2}} + kP = \left(1 - \frac{\sqrt{1-K_1^2}k_{dq} - \sqrt{1-k_{dq}^2}K_1}{\sqrt{1-K_1^2k_{dq}}}\eta\right)\frac{E^2}{2X_S} \tag{22}$$

Therefore, the maximum load active power $P_{\max}$ that the system can carry is related to the penetration ratio of the VSC, as shown in (23):

$$P_{\max} = \frac{E^2}{2X_S}\left(-k + \sqrt{k^2 - \left(1 - \frac{\sqrt{1-k_{dq}^2}K_1}{\sqrt{1-K_1^2k_{dq}}}\right)^2\eta^2 + 1}\right) \tag{23}$$

According to (23), the variation in the system static voltage stabilization domain can be obtained as shown in Fig. 7. Fig. 7 and (23) show that the system static voltage stability is constrained by the system reactive power reserve as the penetration ratio of the VSC increases, resulting in a contraction of the static voltage stability domain. The static voltage stability domain shrinks faster as the penetration ratio increases.

In Fig. 6(b), two cases can be distinguished: case 1, when the reactive power of the synchronous generator of BUS1 reaches its maximum value and precedes that of the VSC of BUS2; and case 2, when the reactive power of the VSC of BUS2 reaches its

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maximum value and precedes that of the synchronous generator of BUS1.

In case 1, the voltage of BUS2 can be maintained constant until the reactive power reserve of the VSC of BUS2 is depleted. The limit-induced bifurcation occurs immediately after the reactive power reserve of BUS2 is exhausted, resulting in system voltage collapse. Therefore, there is also no need to consider the effect of the current limitation. The variation in the system static voltage stability domain with increasing penetration ratio is similar to that in Fig. 6 and will not be discussed in detail here.

In case 2, the VSC can be equated to a current source at the LIB. Neglecting the transverse component in the voltage drops, there is a relationship between U_{pcc} and E:

$$E - \frac{Q_{\mathrm{BUS1}} X_{S1}}{E} = V \tag{24}$$

$$U_{pcc} - \frac{Q_{\mathrm{BUS2}} X_{S2}}{U_{pcc}} = V \tag{25}$$

where Q_{BUS1} is the reactive power of BUS1; Q_{BUS2} is the reactive power of BUS2. It is assumed that the maximum reactive power of the synchronous generator is $(1\text{-}\eta)E^2/2X_{S1}$, so at LIB, it is known that Q_{BUS1} and Q_{BUS2} are as follows:

$$Q_{\mathrm{BUS1}} = \frac{(1-\eta)E^2}{2X_{S1}} \tag{26}$$

$$Q_{\mathrm{BUS2}} = \sqrt{1-k_{dq}^2}\, U_{pcc} I_{\max} \tag{27}$$

where Q_{BUS1} is the reactive power of BUS1; Q_{BUS2} is the reactive power of BUS2. Bringing (26) and (27) into (24) and (25), it can be obtained that at the LIB, U_{pcc} is as follows:

$$U_{pcc} = \frac{(1+\eta)E}{2} + \sqrt{1-k_{dq}^2} I_{\max} X_{S2} \tag{28}$$

The line-consumed and load-consumed reactive power are equal to $Q_{\mathrm{BUS1}}+Q_{\mathrm{BUS2}}$:

$$kP + Q_{S1} + Q_{S2} = \sqrt{1-k_{dq}^2} U_{pcc} I_{\max} + \frac{(1-\eta)E^2}{2X_{S1}} \tag{29}$$

where Q_{S1} is the reactive power consumed by reactance X_{S1}, and Q_{S2} is the reactive power consumed by reactance X_{S2}. I_{G}, the current dispatched by the synchronous generator, is shown in (30):

$$I_{\mathrm{G}} = \frac{\sqrt{(1-\eta)^2 P^2 + Q_{\mathrm{BUS1}}^2}}{E} \tag{30}$$

Therefore, at the LIB, Q_{S1} and Q_{S2} are as follows:

$$Q_{S1} = I_{\mathrm{G}}^2 X_{S1} = (1-\eta)^2 \left(\frac{P^2 X_{S1}}{E^2} + \frac{E^2}{4X_{S1}}\right) \tag{31}$$

$$Q_{S2} = I_{\max}^2 X_{S2} \tag{32}$$

Substituting Q_{S1} and Q_{S2} into (29), the maximum active power of the load can thus be obtained as a function of the power factor and penetration ratio of the VSC:

$$P_{\max} = \frac{-k + \sqrt{k^2 - 4(1-\eta)^2 \frac{X_{S1}}{E^2} c}}{2(1-\eta)^2 X_{S1}} E^2 \tag{33}$$

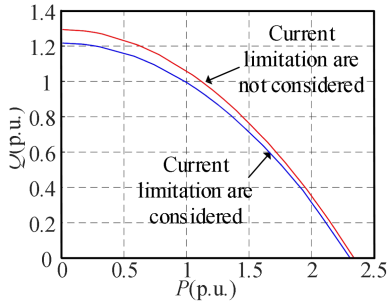


Fig. 8. The static voltage stability domain changes after the influence of the current limitation on the PQ plane is considered.

$$c = I_{\max}^2 X_{S2} + \frac{(\eta^2-1)E^2}{4X_{S1}} - \sqrt{1-k_{dq}^2} U I_{\max} \tag{34}$$

According to (33)–(34), the system static voltage stabilization domain, considering the effect of the current limitation with penetration ratio is 40%, is shown in Fig. 8.

As shown in Fig. 8, Q_{BUS2} is affected by U_{pcc} due to the presence of current limitation, and the static voltage stability domain shrinks by 3.76% on average at a penetration ratio of 40%. A synchronous generator cannot maintain its output at maximum reactive power after the reactive power has been maximized. At this time, the excitation current of the synchronous generator remains constant, so the generator is equivalent to the synchronous reactance before connected to a constant voltage source. The effect of this characteristic on static voltage stability is analyzed in detail in Appendix A, and considered in the complex system.

Notably, the static voltage stability analysis of a single grid-connected VSC system is used to illustrate the research implications of this paper, showing that the influence of reactive power regulation range and the current limitation of VSCs on static voltage stability cannot be ignored. And it is not the focus and innovation of the research in this paper. It is used to analyze the effect of the control characteristics and related elements on the static voltage stability. In a real multi-line power system, no analytical solution exists. So, in the subsequent simulation, the algebraic iterative equations are used for the solution.

IV. Static Voltage Stability Enhancement Method for Power Systems With a High Proportion of VSCs

In the multi-machine system shown in Fig. 9, the reactive power weights [32], [33] and participation factor [41], [42] are usually used to measure the importance of the generator reactive power reserve. The reactive power reserve optimization model is used to improve the system static voltage stability. The reactive power weights λ_{ij} are as follows:

$$\lambda_{ij} = \frac{s_i}{\max(S_j)} \tag{35}$$

$$S = \left[\frac{\partial Q_{\mathrm{G}}}{\partial V_L}\right] \left[\frac{\partial Q_L}{\partial V_L}\right]^{-1} \tag{36}$$

where S_j is the set of the sensitivity matrix at the jth load bus, which is also the jth row of the matrix S; $\max(S_j)$ is the maximum norm of S_j; s_i is the sensitivity value of the ith generator

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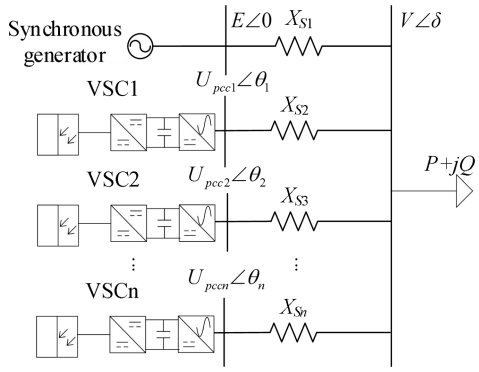


Fig. 9. Diagram of a power system with multiple grid-connected VSCs.

with respect to the jth load bus; $\partial Q_{\text{G}}/\partial V_L$ is the sensitivity of the injected reactive powers of the generator bus with respect to the voltage of the load bus; and $\partial Q_L/\partial V_L$ is the sensitivity of the reactive powers of the load bus with respect to the voltage of the load bus.

The participation factor W_{ij} is shown in (37):

$$W_{ij} = \frac{\Delta Q_{\text{G}i}/\Delta Q_{Lj}}{\max(\Delta Q_{\text{G}}/\Delta Q_{Lj})} \tag{37}$$

where $\Delta Q_{\text{G}i}$ is the reactive power supplied by the ith generator with respect to the reactive power variation of the jth load bus; ΔQ_{Lj} is the reactive power variation of the jth load bus.

A. Reactive Power Support Sensitivity Adapted to the PQ Regulation Characteristics of the VSC

Considering the current limitation characteristics of the VSC, this paper measures the value of the reactive power reserve in terms of the role of the reactive power reserve in maintaining the static voltage stability of the system and the bus voltage immunity, thus defining the reactive power support sensitivity $\alpha_{\text{G}i}$ of the *i*th generator bus as:

$$\alpha_{\text{G}ij} = \frac{\partial V_{\text{G}i}}{\partial Q_j}\frac{\partial Q_{\text{G}i}}{\partial Q_j} \tag{38}$$

$$\alpha_{\text{G}i} = \frac{\alpha_{\text{G}ij}}{\max(\alpha_{\text{G}1j},\ldots\alpha_{\text{G}ij},\ldots a_{\text{G}nj})} \tag{39}$$

where $\partial V_{\text{G}i}/\partial Q_j$ is the sensitivity of the voltage of the *i*th generator bus via the absorbed reactive powers of the *j*th key load bus, and $\partial Q_{\text{G}i}/\partial Q_j$ is the sensitivity of the injected reactive powers of the *i*th generator bus via the absorbed reactive powers of the *j*th key load bus. $\alpha_{\text{G}ij}$ is the sensitivity value of the *i*th generator with respect to the *j*th key load bus, and max ($\alpha_{\text{G}1j}$, $\alpha_{\text{G}ij}\ldots$, $\alpha_{\text{G}nj}$) is the maximum norm of all reactive power support sensitivities.

Since the impedance modulus margin index (IMMI) can accurately assess the level of voltage stability at load buses [43], this paper adopts the IMMI to screen the key load bus and defines μ_j, the IMMI of the *j*th load bus, as:

$$\mu_j = \frac{|Z_{\text{jLD}}| - |Z_{\text{jTHEV}}|}{|Z_{\text{jLD}}|} \tag{40}$$

where $Z_{j\text{LD}}$ is the load impedance of the *j*th load bus, and $Z_{j\text{THEV}}$ is the system's Thevenin equivalent impedance for the *j*th load bus. $\mu_j = 1$ indicates that the bus is highly stable, and $\mu_j = 0$ indicates that the bus is near voltage collapse. Owing to the page limit, the detail derivation of the Thevenin's equivalent impedance seen by the load bus will not be discussed here, which can also be seen in [44], [45].

Therefore, the appropriate value for the IMMI is as large as possible. IMMI only analyzes the current operational status and is equal to the eigenvalue analysis of the Jacobian matrix. The difference is that we can clearly see the degree of weakness of each bus via the IMMI. According to the value of IMMI, the bus voltage stability strength can be sorted, and the voltage stability can be analyzed very intuitively to determine the bus with the weakest system voltage stability.

B. Static Voltage Stability Enhancement Method Based on Reactive Power Reserve Optimization

In this work, the following reactive power reserve optimization model is established to improve the static voltage stability of a system with a high proportion of VSCs. The reactive power reserve optimization model takes the maximum effective reactive power reserve as the objective function, and according to the definition in [33], the effective reactive power reserve is shown in (41):

$$Q_{\text{GR},i} = Q_{\text{Gc},i} - Q_{\text{G0},i} \tag{41}$$

where $Q_{\text{GR},i}$ is the reactive power reserve of the *i*th generator with respect to the voltage collapse, $Q_{\text{Gc},i}$ is the maximum reactive power of the *i*th generator at the point of the voltage collapse, and $Q_{\text{G0},i}$ is the current reactive power dispatched by the *i*th generator. The objective function can be calculated as:

$$\max Q_{\text{GR}} = \max \sum_{i\in N_{\text{G}}} \alpha_{\text{G}i}(Q_{\text{Gc},i} - Q_{\text{G0},i}) \tag{42}$$

where N_{G} is the set of system generator buses. $\alpha_{\text{G}i}$ in the objective function is the reactive power support sensitivity of the *i*th generator bus. The larger $\alpha_{\text{G}i}$ is, the stronger the ability of the reactive power reserve of the *i*th generator bus to support system voltage stability, and the greater the value of its reactive power reserve for voltage stability.

The constraints of the effective reactive power reserve optimization model are as follows:

$$f(\boldsymbol{V},\boldsymbol{\theta},\boldsymbol{P}_l,\boldsymbol{Q}_l,\boldsymbol{P}_{\text{G0}},\boldsymbol{Q}_{\text{G0}},\boldsymbol{I}_d^*,\boldsymbol{I}_q^*,\boldsymbol{u}) = 0 \tag{43}$$

$$\boldsymbol{V}_{\min} \le \boldsymbol{V} \le \boldsymbol{V}_{\max} \tag{44}$$

$$\boldsymbol{Q}_{\text{G,min}} \le \boldsymbol{Q}_{\text{G0}} \le \boldsymbol{Q}_{\text{G,max}} \tag{45}$$

$$\boldsymbol{u}_{\min} \le \boldsymbol{u} \le \boldsymbol{u}_{\max} \tag{46}$$

$$0 \le \boldsymbol{I}_d^* \le \boldsymbol{I}_{d\max} \tag{47}$$

$$0 \le \boldsymbol{I}_q^* \le \boldsymbol{I}_{q\max} \tag{48}$$

where f is the power flow equation; $\boldsymbol{V}$ represents the vector of bus voltage magnitudes; θ represents the vector of bus voltage angles; $\boldsymbol{P}_l$ and $\boldsymbol{Q}_l$ are vectors consisting of active and reactive loads at all buses, respectively; $\boldsymbol{u}$ is the vector of generator

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terminal voltages; and the subscripts max and min represent the upper and lower limits, respectively.

The essence of the proposed method is summarized as follows: 1) The defined reactive power support sensitivity considers the ability to maintain static voltage stability and bus voltage immunity. By introducing the reactive power support sensitivity in the objective function, the PQ current source regulation characteristics of the VSC are considered. 2) It optimizes the $Q_{\mathrm{G}0}$ of each generator bus by solving the voltage of each generator bus. The generator bus with a large reactive power support sensitivity will retain a more effective reactive power reserve and will be able to enhance the static voltage stability more effectively for the power system with a high proportion of VSCs.

C. The Improved Continuation Power Flow for Better Adaptation to Power Systems With a High Proportion of Voltage Source Converters

The continuation power flow (CPF) [46] is used to calculate the maximum load active power that the system can carry under different penetration ratios. During the continuation power flow, the reformulated set of equations remains well conditioned so that divergence and error due to a singular Jacobian are not encountered. As a result, single-precision computations can be used to obtain power flow solutions at and near the critical point.

$$f(\mathbf{V},\boldsymbol{\theta})+\lambda\mathbf{b}=0 \tag{49}$$

where $\boldsymbol{b}$ represents the vector of the generator and the load power change mode, including the direction and magnitude. λ represents the load parameter. CPF is easy to implement, and has a high computational accuracy. So that CPF is a wider application and more recognized method to calculate voltage stability boundary [47], [48]. Moreover, CPF is often used as an exact solution to calibrate the accuracy of the proposed available methods in [49], [50]. Therefore, the CPF is selected to calculate the maximum load active power in this paper.

However, in traditional CPF, when a synchronous generator reactive power reaches the maximum value, the synchronous generator node is usually transformed from a PV bus to a PQ bus. It is therefore necessary to improve the CPF to make it more applicable to the power systems with a high proportion of VSC. Synchronous generator excitation current limitation and VSC current limitation are dynamic processes. Considering their influence in CPF, it is necessary to combine the reactive power characteristics of synchronous generator excitation current limitation and VSC current limitation with the CPF from the steady-state condition. This paper proposes an improved CPF that considers synchronous generator excitation current limitation and VSC current limitation. The flow is shown in Fig. 10.

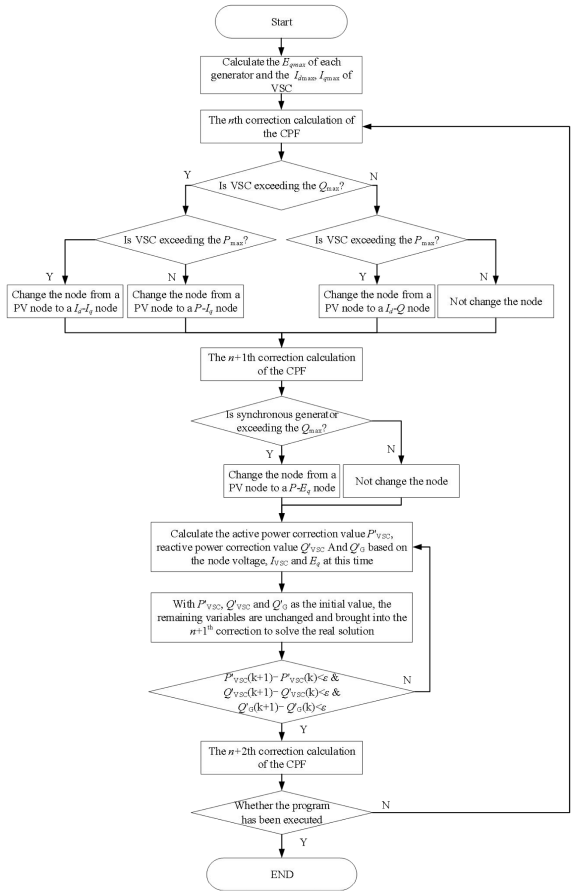


Fig. 10. Improved CPF considering synchronous generator excitation current limitation and VSC current limitation.

1) Before CPF calculation, calculate the maximum internal potential $E_{q\max}$ corresponding to the maximum long-term maintained excitation current of each synchronous generator, the maximum active current $I_{d\max}$ and the maximum reactive current $I_{q\max}$ of each VSC. If the reactive power output $Q_{\mathrm{G}i}$ of the ith synchronous generator exceeds $Q_{\mathrm{G}i\max}$ after the end of the nth correction, the synchronous generator node is converted to P-E_q node. If the active power output $P_{\mathrm{VSC}i}$ exceeds $P_{\mathrm{VSC}i\max}$ or reactive power output $Q_{\mathrm{VSC}i}$ exceeds $Q_{\mathrm{VSC}i\max}$ after the end of the nth correction, the point of common coupling of VSC is converted to P-I_q node or I_d-Q node or I_d-I_q node.
2) Based on the voltage V_i after n+1 correction and the maximum internal potential $E_{q\max}$, the reactive power output correction value $Q'_{\mathrm{G}i}$ can be obtained. And based on the voltage U_{pcci} after n+1 correction of points of common coupling and maximum active current $I_{d\max}$ and maximum reactive current $I_{q\max}$, the reactive power output correction value $P'_{\mathrm{VSC}i}$ and the reactive power output correction value $Q'_{\mathrm{VSC}i}$ can be obtained.
3) $Q'_{\mathrm{G}i}$, $P'_{\mathrm{VSC}i}$ and $Q'_{\mathrm{VSC}i}$ are taken as the parameter of the reference equation, and the n+1 correction is iteratively solved until convergence is achieved. Thus, the actual reactive power output $Q'_{\mathrm{G}i}$ considering the excitation current limitation could be obtained. $P'_{\mathrm{VSC}i}$ and $Q'_{\mathrm{VSC}i}$ considering current limitation could also be obtained.
4) If the power output of other nodes is limited, the same measures shall be taken to iteratively solve the actual power output.

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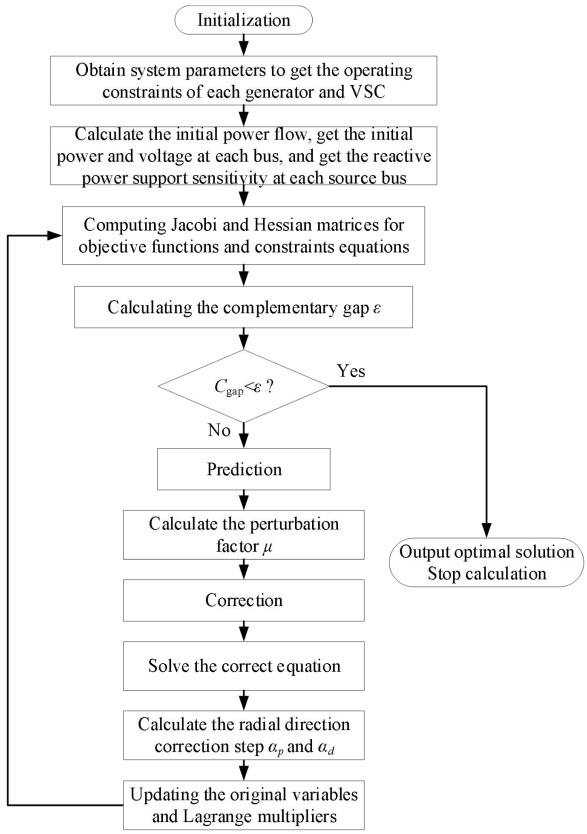


Fig. 11. Computation of the reactive power reserve optimization based on the interior point method.

D. Solving Algorithms

The proposed problem is modeled as a nonlinear optimal reactive power dispatch problem that contains only continuation variables. The dimensionality of the optimization problem depends on the number of reactive power resources. The interior point method always searches for the optimal solution inside the feasible region of the solution, so it can effectively avoid the situation in which the optimal solution falls on the "spur point" of the boundary surface and the algorithm fails to converge. The flow of the reactive power reserve optimization algorithm based on the interior point method is shown in Fig. 11. When the reactive power reserve optimization model constructed via the traditional method and the method proposed in this paper are solved, both methods adopt the interior point method as the solution algorithm. The improvement effect is evaluated by the same evaluation index, which is $P_{\max}$. The only difference is the objective function. The principles of the control variable approach ensure that the results of the comparative test are sufficiently fair.

VI. Simulation and Analysis

A. Simulation Verification in a Test System With Multiple Grid-Connected VSCs

Taking Fig. 9 as a simulation example, where n is taken as 3, the detailed system parameters are shown in Table I. To compare and analyze the difference between a large centralized renewable energy plant and a small renewable energy plant, the capacity ratio of the VSC is chosen as 0.1:0.1:0.8. X_S, X_{S1}, and X_{S2} in Table I refer to the reactance values between the generator nodes and the system in the IEEE-9 bus system in Matpower 4.1 [51]. In a large-scale centralized renewable energy power station, each power generation unit is gathered by long-distance convergence line, and then centrally connected to the grid through transformers. Their grid-connected reactance is generally 3-5 times the grid-connected reactance of synchronous generators [20]. Therefore, the value of X_{S3} is 4.3 times that of X_S.

TABLE I
The Test System Parameters and Variables

Parameters	value	Parameters	value	Variables	Initial value
X_S	0.065 ohm	S_G	(1-η)*6800MVA	E	1 p.u.
X_{S1}	0.065 ohm	S_{VSC1}	0.1η*6800MVA	U_1	1 p.u.
X_{S2}	0.065 ohm	S_{VSC2}	0.1η*6800MVA	U_2	1 p.u.
X_{S3}	0.281 ohm	S_{VSC3}	0.8η*6800MVA	U_3	1 p.u.

TABLE II
The Comparison of the Optimization Results of P_{MAX} When K is 0

Penetration ratio	P_{max0}(MW)	P_{max1}(MW)/ percent load growth (%)	P_{max2}(MW)/ percent load growth (%)	P_{max3}(MW)/ percent load growth (%)
40%	1001.41	1065.72/6.42	1038.56/3.71	1037.2/3.57
50%	816.51	884.89/8.37	853.99/4.59	852.64/4.42
60%	685.85	743.95/8.47	713.64/4.05	712.52/3.89
70%	590.07	640.17/8.49	612.95/3.88	611.97/3.71
80%	517.14	561.07/8.49	537.28/3.89	536.37/3.72
90%	460.03	499.17/8.51	477.26/3.75	476.5/3.58

In addition, this paper assumes the following:

1) The load increases with a constant power factor.
2) The active power dispatched by the synchronous generator and VSC increases in equal proportions to the initial benchmark value.
3) When the reactive power of the synchronous generator or VSC exceeds the limit, the synchronous generator rated excitation current is the maximum excitation current that can be maintained for a long time. It is assumed that the excitation limit is controlled by integration, and that terminal voltage does not affect the limiting ability of the excitation current. The VSC adopts the dq-axis current proportional limitation method, and it is assumed that the voltage reduction at the point of common coupling will not affect the limiting ability of the current limitation.
4) When the synchronous generator is replaced by the VSC, the reactive power reserve of the replacement bus will decrease from the original Q_G to $Q'_G = (1-0.39\eta)\ Q_G$ for $k_{dq} = 0.909$ and $K_1 = 0.8$.

A comparison of the optimization result is shown in Fig. 12 and Table II. In Table II, $P_{\max 0}$ is $P_{\max}$ before optimization, $P_{\max 1}$ is $P_{\max}$ after optimization via reactive support power

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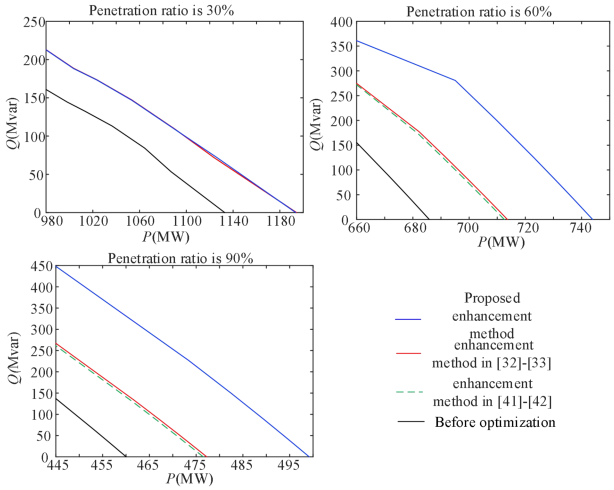


Fig. 12. Comparative results of different static voltage stability enhancement methods.

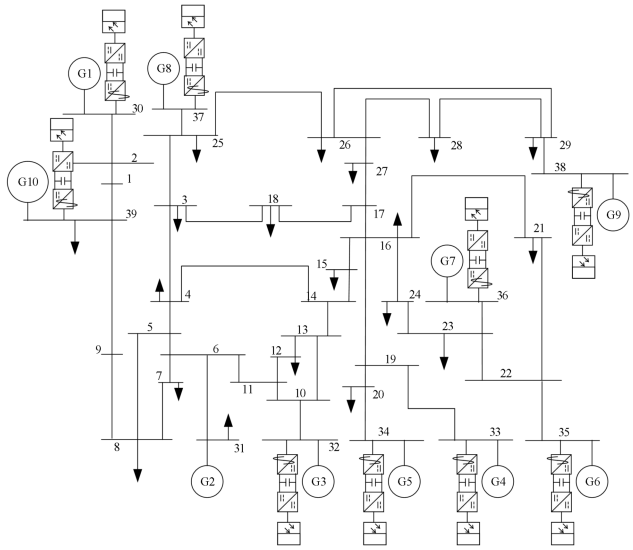


Fig. 13. An extended IEEE-39 bus system with a high proportion of VSCs.

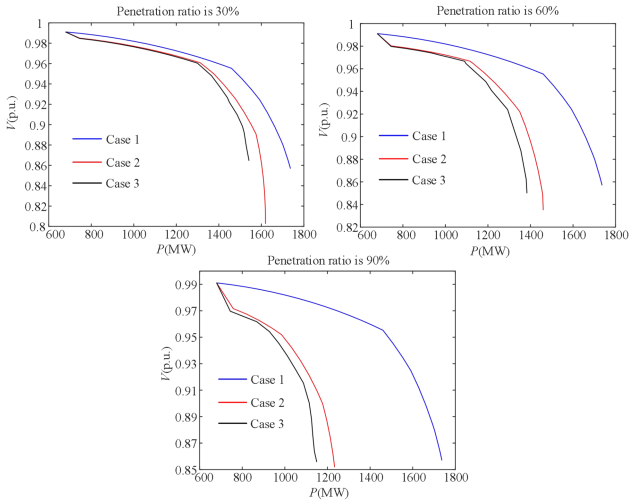


Fig. 14. The P-V curves of key load buses tracked under different penetration ratios of VSCs.

sensitivity, and $P_{\max 2}$ is $P_{\max}$ after optimization via the reactive power weight. $P_{\max 3}$ is $P_{\max}$ after optimization via the participation factor. In Fig. 11, three more representative penetration ratios are selected (30%, 60% and 90%) to increase the legibility and appreciation.

Fig. 12 and Table II show that at a low VSC penetration ratio ($\leq$30%), the static voltage stability domains obtained by optimizing the effective reactive power reserve are consistent when either the reactive power weights or the reactive power support sensitivity proposed in this paper are used. However, at medium VSC penetration ratios (40%-60%) and high VSC penetration ratios (>60%), the voltage stability domain obtained by optimizing the reactive power reserve using the reactive power support sensitivity proposed in this paper will be larger than that of the conventional method. Taking the penetration ratio of 40% and the load power growth direction k is 0 as an example, by adopting the static voltage stability enhancement method proposed in this paper, the maximum load active power that the system can carry grows from 1001.41 MW before optimization to 1065.72 MW, which is a growth of 6.42%. However, $P_{\max 2}$ is only 1038.56 MW, which is a growth of 3.71%. $P_{\max 3}$ is only 1037.2 MW, which is a growth of 3.57%. The $P_{\max}$ optimized by the proposed method is much greater than that of the traditional method, and the higher the penetration ratio is, the better the optimization results of static voltage stability.

B. Quantitative Analysis of the Current Limitation and Reactive Power Regulation Ability of the VSCs

The detailed system parameters of the extended IEEE-39 bus system with a high proportion of VSCs (Fig. 13) are described in the MATPOWER 4.1 [47]. The IEEE 39-bus test system contains 10 generator buses. Generator bus 31 is a slack bus, and its reactive power regulation capacity is infinite. The assumption is the same as that in the simulation above. By setting the penetration ratio of VSCs to increase from 0, this paper studies the influence of the gradual replacement of the synchronous generator by the VSC on the static voltage stability. The IMMI of each load bus is shown in Table VI of Appendix B. As shown in Table VI, the IMMI of load bus 20 is the smallest, so load bus 20 is selected as the key load bus. The following three cases are considered:

Case 1: There is no grid-connected VSC.
Case 2: The current limitation is not considered with the grid-connected VSC.
Case 3: The current limitation is considered with a grid-connected VSC.

A comparison of the P-V curves of key load buses tracked under different penetration ratios of VSCs is shown in Fig. 14.

In Fig. 14, at a low VSC penetration ratio ($\leq$30%), there is little difference in the P-V curves of the key load bus among the three cases. At medium VSC penetration ratios (40%$\sim$60%) and high VSC penetration ratios (>60%), the ability of the VSC PQ regulation characteristics to influence the static voltage stability is significant, so the P-V curves of Case 2 and Case 3 have

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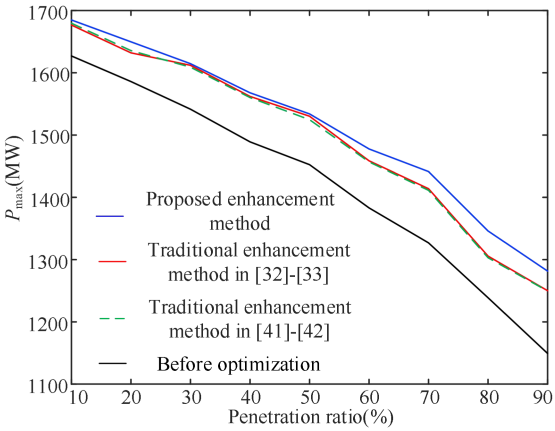


Fig. 15. Comparison of the optimization results of the static voltage stability under different penetration ratios.

a significant tendency toward inward contraction. The higher the penetration ratio is, the greater the influence of the current limitation, and the greater the difference between the P-V curves of Case 3 and Case 2. Case 2 which does not consider the current limitation in VSCs yields more optimistic results. In Case 3, the "moment" when each synchronous generator and VSC reach the reactive power output limit is greater than that in Cases 1 and 2, and the number of source buses reaching the reactive power output limit increases, which ultimately leads to further inward folding of the P-V curve of the system and further degradation of the static voltage stability.

The initial voltage drop (at the start of the loading) increases because the reactive power reserve at the generator nodes decreases gradually with increasing penetration ratio. When the P-V curve is plotted, the reactive power of some generator buses in the system reaches its maximum reactive power as soon as the load starts to increase. With the larger the penetration ratio is, the more generator buses reach the maximum reactive power at the start of the loading, which increases the initial voltage drop (at the start of the loading).

For example, penetration ratios of 30%, 60% and 90% as used. When penetration ratio is 30%, BUS34 reaches maximum reactive power at the start of the loading. When penetration ratio is 60%, BUS32 and BUS34 reach maximum reactive power at the start of the loading. When penetration ratio is 90%, BUS32, BUS33, BUS34 and BUS35 reach maximum reactive power at the start of the loading.

C. Comparative Analysis of the Static Voltage Stability Enhancement Methods

To verify the effectiveness of the proposed enhancement method in this paper, it is compared with the conventional reactive power reserve optimization method using the reactive power weights proposed in [32], [33] and participation factor in [41], [42]. The comparative results of the static voltage stability enhancement under different penetration ratios can be obtained, as shown in Fig. 15 and Table III.

Fig. 15 and Table III show that $P_{\max}$ is significantly increased, and the static voltage stability is effectively improved. Using the reactive power support sensitivity proposed in this paper to optimize the system reactive power reserve, at high penetration ratio, the proposed methodology has a lift percentage of 9.60% compared with 6.76% for the conventional method in [32], [33], and 6.91% for the conventional method in [41], [42], The proposed methodology has 2.84% and 2.69% more lift than the conventional method.

The static voltage stability enhancement method based on reactive power reserve optimization proposed in this paper considers the influence of the current limitation, which can better distinguish the reactive power value of each source bus in a complex power system with a high proportion of VSCs. Such that the system static voltage stability can be enhanced more effectively.

TABLE III
COMPARISON OF THE OPTIMIZATION RESULTS OF P_{MAX} WHEN K IS 0

Penetration ratio	P_{max0}(MW)	P_{max1}(MW)/ percent load growth (%)	P_{max2}(MW)/ percent load growth (%)	P_{max3}(MW)/ percent load growth (%)
10%	1626.98	1684.85/3.56	1679.61/3.23	1676.67/3.05
20%	1585.96	1649.8/4.02	1635.64/3.13	1632.03/2.9
30%	1541.51	1614.66/4.75	1609.07/4.38	1612.01/4.57
40%	1489.02	1567.83/5.29	1560.23/4.78	1562.08/4.91
50%	1452.38	1534/5.62	1524.7/4.98	1530.3/5.36
60%	1383.12	1477.7/6.84	1456.94/5.34	1458.64/5.46
70%	1326.91	1441.42/8.63	1411.28/6.36	1413.95/6.56
80%	1238.81	1345.99/8.65	1303.12/5.19	1305.7/5.4
90%	1149.27	1281.51/11.51	1249.52/8.72	1250.02/8.77

TABLE IV
THE COMPUTATIONAL TIME COMPARED IN THE SYSTEM WITH MULTIPLE GRID-CONNECTED VSCS

Method	t_1 (in sec)	t_2 (in sec)	t_3 (in sec)
The method in [32]-[33]	0.15	0.18	72.2
The method in [41]-[42]	0.16	0.17	72.5
The proposed method	0.15	0.19	72.8

TABLE V
THE COMPUTATIONAL TIME COMPARED IN IEEE-39 BUS SYSTEM

Method	t_1 (in sec)	t_2 (in sec)	t_3 (in sec)
The method in [32]-[33]	0.25	9.47	427.4
The method in [41]-[42]	0.24	9.52	428.7
The proposed method	0.25	9.54	426.1

D. Performance of Computational Efficiency

In this section, the computational efficiency of the proposed method is investigated. The computation time is dependent on the reactive power reserve optimization and CPF. CPF is conducted by using a computer with an Intel Core i7-10875H CPU at 2.30 GHz and 16.0 GB of RAM running MATLAB 2023a. The results are shown in Tables IV and V. t_1 is the computational time of the index, t_2 is the computational time of solving the optimization mode, and t_3 is the computational time of CPF.

With an increase in the number of reactive power resources, the total computation time increases slightly. The method

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proposed in this paper is the same as the static voltage stabilization enhancement model constructed via the traditional method in terms of computational complexity and computational solvability. The time required for solving the optimization model is 9.54 s, which is still acceptable because the time period of reactive power reserve optimization in hard real-time systems is usually 15 minutes. However, real-time sensing and highly accurate prediction are needed to ensure that the system maintains changes in the system state.

V. Conclusion

In this paper, the static voltage stability enhancement method for a power system with a high proportion of VSCs is studied. In a single grid-connected VSC system, this paper first analyzes the PQ regulation characteristics of the VSC and synchronous generator and establishes an analytical model of static voltage stability by considering the current limitation. The analytical results show that when the penetration ratio of the VSC reached 40%, the system static voltage stability domain would be reduced by 3.76% on average.

By determining the current voltage and effective reactive power reserve of each source bus according to the reactive power support sensitivity, the proposed method can effectively improve the static voltage stability. Two simulation cases have shown that at medium and high VSC penetration ratios (≥40%), compared with the traditional method, the static voltage stability enhancement method proposed in this paper is able to better improve the static voltage stability of a system with a high proportion of VSCs. In the test system, at penetration ratio of 40%, the proposed methodology has 2.71% and 2.85% more lift than the conventional method, respectively. In the extended IEEE-39 bus system, at high penetration ratio, the proposed methodology has 2.84% and 2.69% more lift than the conventional method does.

However, this paper has not yet considered the intermittence and variability of VSC resource power output. Non-continuous variable reactive power compensation devices such as casting capacitors are not considered. In future work, the model will be transformed into a mixed-integer linear programming model. And an intraday reactive power reserve optimization method can be explored for adoption in hard real-time systems.

Appendix A

To simplify the calculation, considering the generator as a non-saturated hidden pole machine and neglecting the armature resistance, the synchronous reactance of the d- and q-axes can be expressed as $X_d = X_q = X$. Fig. 16 shows the synchronizing reactance of the generator on the basis of Fig. 6.

The relationships between the active and reactive powers of the synchronous generators P_{G} and Q_{G} and the synchronous reactance X of the synchronous generator are shown in (A1)-(A2):

$$P_{\mathrm{G}} = \frac{E_q E}{X}\sin(\delta_2 - \delta) \tag{A1}$$

$$Q_{\mathrm{G}} = \frac{E_q^2 - E_q E\cos(\delta_2 - \delta)}{X} \tag{A2}$$

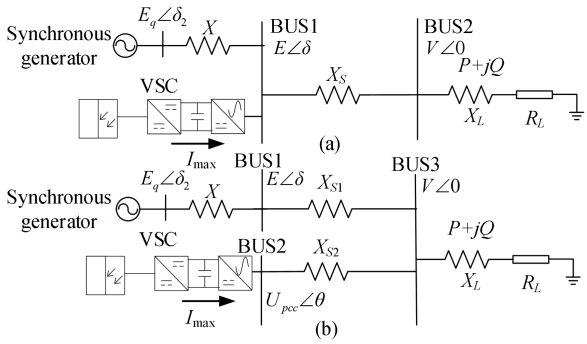


Fig. 16. Diagram of a single VSC grid-connected system considering synchronous reactance.

where E_q is the no-load electromotive force and δ_2 is the power angle of the no-load electromotive force. There is a relationship between E_q and the excitation current i_f:

$$E_q = x_{ad} i_f = \omega_0 L_{ad} i_f \tag{A3}$$

where x_{ad} is the straight-axis armature reactance, ω_0 is the rated angular frequency, and L_{ad} is the mutual inductance coefficient between the excitation winding and the straight-axis equivalent winding.

When the synchronous generator is running steadily at the rated excitation current i_{fN}, E_q remains constant. On the basis of (A1)–(A3), the analysis of the static voltage stability of Fig. 16 can be carried out in accordance with the analysis in Section III.

Appendix B

The IMMI of each load bus is shown in Table VI.

TABLE VI
IMMI of Each Load Bus in IEEE-39 Bus System

Bus	IMMI	Bus	IMMI
1	0.983	20	0.920
3	0.966	21	0.964
4	0.940	23	0.976
7	0.966	24	0.964
8	0.928	25	0.980
9	0.989	26	0.980
12	0.972	27	0.953
15	0.954	28	0.959
16	0.974	29	0.967
18	0.980		

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